FARUMASI Pharmacist Assistant: A Grounded Domain-Specific Question-Answering System for Community Pharmacy in Rwanda

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Summative Project — Natural Language Processing (Sep26)
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GitHub: https://github.com/Emmanuel-NS/FARUMASI-AI
Live app: https://huggingface.co/spaces/Emmanuel-NS/farumasi-pharmacist-qa

Abstract

Patients in Rwanda often use community pharmacies as a first point of care, yet reliable, plain-language answers about medicines are hard to obtain online, and general-purpose chatbots can invent doses. This project builds a **domain-specific English question-answering (QA) system** for routine medicine and minor-ailment questions, grounded only in trusted passages (US openFDA labels for WHO essential medicines, MedQuAD, MedicationQA, and Rwanda FDA patient leaflets). The pipeline combines a DistilBERT intent classifier, a fine-tuned MiniLM dense retriever, and a LoRA-adapted Qwen2.5-1.5B-Instruct chat reader (with a Flan-T5-base + LoRA fallback), plus rule-based safety and query understanding. Systematic experiments against TF-IDF, BM25, zero-shot Transformers, and extractive baselines show that LoRA fine-tuning roughly doubles ROUGE-L over zero-shot Qwen (0.327 → 0.643 with gold context) while raising passage-groundedness (support) from 0.62 to 0.96 and abstention on unrelated passages from 25% to 73%. The system is deployed as a Gradio web app on Hugging Face Spaces (ZeroGPU). This report documents the problem, data, methodology, experiments, evaluation, error analysis, deployment, and limitations.

Keywords: domain-specific QA; retrieval-augmented generation; LoRA; biomedical NLP; pharmacy; Rwanda; DistilBERT; MiniLM; Qwen2.5; Flan-T5

1. Introduction and Problem Definition

Community pharmacies in Rwanda are a primary access point for medicine advice, but pharmacists are time-constrained and online information is fragmented between US brand names, clinician-facing labels, and informal social media advice. A general chatbot is unsuitable for this setting: large language models (LLMs) can hallucinate doses, interactions, and contraindications [1], [2]. The research problem addressed here is therefore:

How can we build an English-language, domain-specific QA assistant that answers routine medicine and minor-ailment questions for Rwandan patients using only trusted leaflet text, refuses out-of-scope queries, and flags risky situations—without relying on an external commercial chatbot API?

1.1 Task, language, and target users

Task. Extractive / generative open-domain QA over a fixed pharmacy corpus (retrieval + grounded answer generation), with an upstream 3-way intent classifier (medicine_info / condition_advice / out_of_domain) and a rule-based risk detector. This is **not** machine translation; language choice is justified by data availability and by the fact that most official medicine labels and NIH health topics used in training are English, while Kinyarwanda/French leaflets are sparse and often scanned images.

Language. English only for this version. Rwandan community pharmacy practice commonly uses English brand and generic names alongside Kinyarwanda speech; the assistant expands local lay terms and brand names (e.g. *panadol* → paracetamol, *coartem* → artemether/lumefantrine, *after sex pills* → levonorgestrel) so English retrieval still serves local phrasing. Multilingual QA is left as future work because half of the Rwanda FDA PDFs lack a text layer.

Target users. Adult patients and caregivers seeking routine information about WHO essential medicines and minor ailments, plus pharmacy students and partners of the FARUMASI digital pharmacy platform. The assistant is explicitly **not** a substitute for a pharmacist or doctor; every answer carries a disclaimer, and emergencies are directed to SAMU (912).

1.2 Research questions

- **RQ1.** Does DistilBERT fine-tuning outperform TF-IDF baselines on informal consumer questions?
- **RQ2.** Does dense MiniLM fine-tuning with in-batch negatives beat BM25 and zero-shot MiniLM on Recall@k / MRR?
- **RQ3.** Does LoRA fine-tuning of a reader (Flan-T5, then Qwen2.5-Instruct) improve grounded answer quality over extractive and zero-shot baselines?
- **RQ4.** Does including Rwanda FDA leaflet questions improve retrieval on local brands, or does US/NIH training already transfer?

2. Related Work

2.1 Biomedical and consumer health QA

Biomedical QA has progressed from IR over curated corpora to neural readers. **MedQuAD** [3] aggregates large NIH/MedlinePlus-style QA collections and frames answering partly as question–entailment; **MedicationQA** [4] focuses specifically on real consumer medicine questions with answers drawn from trusted resources—closer to our target user phrasing than clinician notes. Earlier clinical QA often assumed expert users; consumer health QA must handle lay synonyms, incomplete questions, and higher stakes when the system is wrong [2]. Our contribution is not a new dataset alone, but a **closed-corpus RAG pipeline** adapted to Rwandan pharmacy practice (local brands, emergency contraception lay terms, SAMU referral) on top of these public resources plus Rwanda FDA leaflets [20].

2.2 Retrieval and grounded generation

Dense bi-encoders such as Sentence-BERT and MiniLM [5], [6] map questions and passages into a shared vector space for cosine ranking; classical **BM25** [7] remains strong when rare brand tokens dominate matching—which we observe for Rwanda leaflet names. **Retrieval-augmented generation (RAG)** [11] conditions a generator on retrieved documents to reduce hallucination. Instruction-tuned generators including **Flan-T5** [8] and **Qwen2.5** [9] can be adapted with **LoRA** [10], updating a small set of low-rank matrices instead of all weights—essential on free Colab T4 hardware. Our experiments connect to this literature by showing that (i) domain LoRA roughly doubles zero-shot Qwen ROUGE-L while raising a simple groundedness proxy (support) from 0.62 to 0.96, and (ii) zero-shot instruction models paraphrase freely and under-abstain, consistent with known hallucination risks [1], [2].

2.3 Scope control and safety

Out-of-scope detection for dialogue systems is studied in CLINC150 [13]. We treat scope as a first-class classifier before retrieval so non-health questions never reach the reader. Commercial clinical chatbots and closed APIs are deliberately avoided: the assessment rubric forbids simply wiring an existing chatbot API, and auditable grounding requires that every answer cite passages from a known corpus. Rule-based emergency overlays complement learned modules because safety actions (e.g. “call 912”) must be predictable rather than sampled [12].

3. Dataset

3.1 Sources, size, and characteristics

All training and evaluation text is human-written. No LLM-generated passages were used as gold answers. The passage corpus contains **8,840** passages; the QA set contains **9,518** question–answer pairs grouped by medicine/condition so the same entity does not appear in more than one split (train 7,200 / val 1,136 / test 1,182).

Dataset	Role	Size used	Licence / notes
openFDA drug labels (US FDA)	Medicine passages for ~140 WHO essential medicines	2,049 passages; 2,523 templated QA	Public domain
MedQuAD [3]	Condition / health-topic QA	5,546 passages; 5,580 QA	CC BY 4.0
MedicationQA [4]	Real consumer medicine questions	655 QA	CC BY 4.0
Rwanda FDA Patient Information Leaflets	Local brands (e.g. Coartem, Agycin)	187 usable leaflets → 590 passages; 760 QA	Public regulatory PDFs
CLINC150 [13]	Out-of-domain intent negatives	23,700 queries (sampled)	CC BY 3.0

Dataset	Role	Size used	Licence / notes
ChatDoctor-HealthCare Magic [14]	Weakly labelled consumer questions (intent train only)	3,416 questions	Apache 2.0
Challenge set (this project)	56 informal consumer questions; eval only	56	Author-labelled

Table 1. Datasets used in the project.

3.2 Preprocessing and quality

- Label sections cleaned and split into ≤160-word passages; mean passage length ≈ 118 words.
- openFDA and Rwanda leaflet questions are templated over real section text (documented limitation); MedicationQA and MedQuAD questions are natural.
- Splits are **entity-grouped** to avoid leakage (a medicine in train cannot appear in test).
- Rwanda PDFs: of 416 cached files, 213 scanned images without a text layer were excluded; two-column layouts sometimes interleave columns (noise).
- Supplementary corrected labels for metformin (single ingredient), levonorgestrel (OTC emergency pill), and ulipristal (120-hour emergency pill) replace wrong combination-product downloads.
- Passage cleaning removes cross-references such as "[see Warnings…]" and US MedWatch boilerplate before the reader sees text.

3.3 Dataset limitations

Medicine answers rely heavily on FDA wording written for clinicians; MedQuAD drug answers were removed by their authors for copyright. Templated questions inflate lexical overlap metrics. US brand names and doses may differ from Rwandan practice. English-only coverage excludes many real patient utterances. These limitations motivate grounded generation, honest abstention for unknown local brands, and a strong disclaimer.

4. Methodology

Figure conceptually: question → (optional) small-talk / feedback rules → risk detector → intent classifier → query expansion (lay/brand terms) → dense retriever with medicine/section re-ranking → chat reader (Qwen + LoRA) grounded on on-topic passages → safety localisation (e.g. US poison number → 912) + disclaimer + sources.

4.1 Intent / scope classifier

Architecture. DistilBERT [15] (6-layer Transformer encoder, distilled from BERT) with a linear classification head over three labels. Input: tokenised question (max length 64). Output: label + softmax confidence. **Baselines.** TF-IDF + logistic regression with word and word+char n-grams. **Training.** Fine-tuned on MedQuAD/CLINC, then on a consumer-augmented set mixing real patient questions (HealthCareMagic) with heuristic labels. **Why DistilBERT.** Course-appropriate Transformer, fits free Colab hardware, strong text classification baseline, and small enough for ZeroGPU alongside retrieval and generation.

4.2 Retriever

Architecture. all-MiniLM-L6-v2 bi-encoder [6]: question and passage mapped to 384-d normalised embeddings; ranking by cosine similarity. **Fine-tuning.** Contrastive training with in-batch negatives (2 epochs, batch 64, lr 2e-5). **Inference.** Top-50 dense hits are re-ranked with boosts when the passage medicine matches a drug named in the query (+0.15) and when the section matches the asked aspect (+0.08). Stub/empty passages are masked. **Baselines.** BM25 [7]; zero-shot MiniLM; hybrid BM25+dense via Reciprocal Rank Fusion (RRF).

4.3 Answer readers (Flan-T5 LoRA and Qwen LoRA)

Flan-T5-base + LoRA. Encoder–decoder instruction model [8]. LoRA adapters (rank 16) on query/value projections (~0.7% of parameters). Trained to produce short answers from a prompt containing three passages (gold + hard negatives). Hyperparameters for the reported run: 2 epochs, lr 3e-4, effective batch 16, ~25 minutes on a Colab T4.

Qwen2.5-1.5B-Instruct + LoRA (deployed reader). Decoder-only chat model [9]. LoRA (r=16, α=32, dropout 0.05) on attention and MLP projections (q/k/v/o, gate/up/down). Chat template with a system prompt forbidding invention; loss masked on prompt tokens (answer tokens only). Three example types: (i) answer with opener +

passage body; (ii) multi-section overview; (iii) ~8% “my sources don’t cover this” with unrelated passages. Hyperparameters: 1 epoch, max_train 3,000, batch 2, grad_accum 8, max_len 1536, lr 2e-4, fp16, gradient checkpointing, Colab T4 (~1–2 h). At inference the model sees on-topic retrieved passages only; overviews are written per section in one batch so headings stay aligned.

4.4 Query understanding and safety (non-learned)

Lay/brand expansion, follow-up resolution (“what do they do?” inherits the previous medicine), progressive overviews, unknown-product detection (e.g. Kopavitamin), fuzzy suggestion of close names (quarteme → Coartem), and rule-based risk levels (routine / prescription / special group / emergency) are implemented in `pharmqa/query.py` and `pharmqa/safety.py`. These components are transparent by design and were refined from live-app failures that benchmarks alone did not expose.

4.5 Training and inference hyperparameters (summary)

Component	Base model	Adaptation	Key hyperparameters	Hardware
Intent	DistilBERT-base-uncased	Full fine-tune + consumer data	max_len 64; 3 labels	Colab T4
Retriever	all-MiniLM-L6-v2	Contrastive fine-tune	2 epochs; batch 64; lr 2e-5; in-batch negatives	Colab T4
Reader (legacy)	Flan-T5-base	LoRA r=16 (q/v)	2 epochs; lr 3e-4; eff. batch 16; max_src 512	Colab T4 ~25 min
Reader (deployed)	Qwen2.5-1.5B-Instruct	LoRA r=16 (attn+MLP)	1 epoch; lr 2e-4; batch 2×8; max_len 1536; answer-token loss	Colab T4 ~1–2 h

Table 1b. Modelling choices used for the reported results and the live app.

5. Baseline and Experimentation

Each component has a meaningful baseline and controlled ablations. Experiments were run on Colab T4 GPUs; metrics are stored under `results/` for reproducibility.

5.1 Intent experiments

Baseline rationale. TF-IDF + logistic regression is a strong classical text classifier and isolates whether Transformers help beyond bag-of-words. **What changed.** (i) word vs word+char n-grams; (ii) DistilBERT without consumer augmentation; (iii) DistilBERT with HealthCareMagic questions. **Why.** Models trained only on formal MedQuAD/CLINC text scored >99% on the held-out test split but collapsed on the informal challenge set—evidence they learned *style* rather than topic. Consumer augmentation tests whether realistic phrasing closes that gap (RQ1).

5.2 Retrieval experiments

Baseline rationale. BM25 is the standard sparse IR baseline; zero-shot MiniLM shows pretrained semantic matching without domain fine-tuning. **What changed.** Fine-tune MiniLM on the pharmacy QA pairs; train with vs without RwandaFDA-PIL questions; fuse BM25 with dense ranks via RRF. **Why.** RQ2–RQ4: measure gains from domain adaptation and whether local leaflets are necessary for local brands.

5.3 Generation experiments

Baselines. Extractive (first sentences of top passage); Flan-T5 zero-shot; Qwen2.5-1.5B zero-shot. **Proposed.** Flan-T5 + LoRA; Qwen2.5 + LoRA. **Context conditions.** *Gold* = gold passage + 2 BM25 hard negatives (isolates the reader); *Retrieved* = top-3 BM25 (end-to-end with a weaker retriever so Flan/Qwen tables remain comparable). **Why.** RQ3: test whether LoRA teaches the model to select the right passage and phrase a grounded answer, and whether a chat-style reader improves openings and abstention.

6. Results

6.1 Intent classification

Model	Test acc	MedicationQA acc	Challenge acc	Challenge macro-F1
TF-IDF + LR (word)	0.993	0.528	0.393	0.372
TF-IDF + LR (word+char)	0.995	0.694	0.464	0.420
TF-IDF + LR + patient Qs	0.979	0.556	0.768	0.764
DistilBERT (original)	0.998	0.917	0.554	0.486
DistilBERT + patient Qs (deployed)	**0.994**	**0.889**	**0.804**	**0.782**

Table 2. Intent results. High formal-test accuracy is misleading; the challenge set is the realistic criterion. Consumer augmentation is the decisive factor.

6.2 Retrieval

Model	R@1	R@5	MRR@10	Topic@1	Rwanda R@1
BM25	0.322	0.706	0.478	0.852	0.444
MiniLM zero-shot	0.436	0.804	0.589	0.882	0.400
MiniLM FT (no Rwanda)	0.694	0.954	0.809	0.977	0.667
MiniLM FT (all, deployed)	**0.695**	**0.958**	**0.810**	**0.977**	**0.678**
Hybrid RRF	0.551	0.867	0.684	0.951	0.689

Table 3. Retrieval on 1,182 test questions / 8,840 passages. Fine-tuning more than doubles BM25 Recall@1. Adding Rwanda questions helps only ~1 point on Rwanda R@1 (within noise for n=180): the model mainly learns question→section mapping that transfers.

6.3 Answer generation

Reader	Context	ROUGE-L	BERTScore	Support	Words	Abstain unrelated
Extractive*	gold	0.931*	0.985*	—	62	no
Flan-T5 zero-shot	gold	0.191	0.761	—	9	no
Flan-T5 + LoRA	gold	0.608	0.900	—	37	no
Qwen zero-shot	gold	0.327	0.829	0.624	58	25%
Qwen + LoRA (deployed)	gold	**0.643**	**0.911**	**0.964**	104	**73%**
Extractive	retrieved	0.395	0.833	—	60	no
Flan-T5 + LoRA	retrieved	0.414	0.841	—	35	no
Qwen + LoRA	retrieved	**0.432**	**0.847**	0.930	99	73%

*Table 4. Generation results. *Extractive on gold is an upper bound (references are the first sentences of the gold passage). Support = share of answer content words found in the given passages (groundedness). Qwen LoRA never falsely abstained when the gold passage was present (false abstention = 0).*

6.4 Safety rules

On the labelled challenge risk set, the rule-based detector achieved accuracy 1.0 across routine, prescription, special-group, and emergency classes. This score is optimistic because the same set helped refine the rules; it is reported for transparency, not as an independent clinical validation.

6.5 Discussion: connecting results to the literature

Three findings align with prior work. First, high in-domain classification accuracy can mask poor generalisation to consumer style—consistent with intent OOS research on CLINC150 [13] and with the need for realistic evaluation beyond clean test splits. Second, BM25's strength on rare brand tokens vs dense models' strength after fine-tuning mirrors classic sparse/dense trade-offs [7], [5]. Hybrid RRF helping only when the dense model is weak matches the intuition that fusion cannot rescue a weaker ranker once the strong ranker already dominates. Third, zero-shot instruction models produce fluent but under-grounded answers (support 0.62), while LoRA on grounded targets restores fidelity—empirically supporting RAG + parameter-efficient adaptation [10], [11] over "prompt the base model and hope". The chat reader's higher abstention rate (73% vs 25%) is particularly important for pharmacy use: refusing is safer than inventing a local brand's ingredients.

7. Evaluation Metrics and Error Analysis

7.1 Why these metrics

- **Accuracy / macro-F1** for intent: classes are imbalanced; macro-F1 penalises ignoring a class.
- **Recall@k and MRR@10** for retrieval: standard IR metrics; Topic@1 checks whether the right medicine is found even if the wrong section is ranked first.
- **ROUGE-L** [16] for lexical overlap with reference answers; we also report **ROUGE-L recall** because chat answers are longer by design.
- **BERTScore** [17] (DistilBERT) for semantic similarity when wording differs.
- **Support** and **abstention rate** for groundedness and safety of "I don't know" behaviour—lexical metrics alone cannot capture hallucination.

None of these metrics measure clinical correctness; they measure agreement with trusted source text and discrimination of scope. That limitation is discussed in Section 9.

7.2 Successful examples (live system)

- **"What are the side effects of metformin?"** → opens with "Metformin can cause side effects." then lists diarrhoea, nausea, etc. from the single-ingredient label (earlier bug: metoclopramide text).
- **"after sex pills" then "any of 120hrs, I passed 72hrs"** → overview of levonorgestrel, then ulipristal dosing within 120 hours.
- **"what do you know about quarteme"** → "closest name is Coartem" + malaria overview (artemether/lumefantrine).
- **"exchange rate of the Rwandan franc"** → low-confidence / out-of-scope refusal (no medicine answer invented).

7.3 Failure modes and explanations

- **Style overfitting (intent).** Without consumer questions, DistilBERT was near-perfect on formal test text but ~55% on the challenge set—misreading casual health wording as out-of-domain.
- **Aspect errors (retrieval).** BM25 Topic@1 is high (0.85) but R@1 is low (0.32): it finds the drug but the wrong leaflet section. Fine-tuning mainly fixes aspect ranking.
- **Wrong openFDA product.** The "richest label" download sometimes selected combination products; silent wrong answers until supplementary labels were added.
- **Unknown local brands.** "Kopavitamin" triggered generic vitamin text until NameChecker forced an honest "product not in sources" reply.
- **Reader opener mismatch.** Qwen sometimes started with a side-effects sentence for a "what is that?" follow-up; a post-hoc opener correction aligns the announced section with the question.
- **Clinician wording.** Even correct answers retain FDA phrasing; symptom questions may explain the condition before naming OTC options (acetaminophen/ibuprofen).
- **Loose abstention.** On deliberately unrelated passages, Qwen LoRA still answers ~27% of the time from loosely related text—hence the retrieval similarity threshold remains a second guard.

8. Web Deployment

The final system is a **Gradio 5** web application hosted on Hugging Face Spaces with ZeroGPU. At startup, DistilBERT, MiniLM embeddings, and the Qwen base model + merged LoRA adapter are loaded on CPU; a `@spaces.GPU` wrapper moves compute to the GPU only for each answer (~5–10 s). The UI shows the chat, an "Understood as" follow-up interpretation, model component tags, and source passages with similarity scores. Deployment artefacts and `scripts/deploy_space.py` reproduce the Space contents. Live URL: https://huggingface.co/spaces/Emmanuel-NS/farumasi-pharmacist-qa

9. Limitations and Ethical Considerations

- English only; Kinyarwanda and French are out of scope for this version.
- Templated openFDA/Rwanda questions and clinician-facing label wording limit ecological validity.
- ROUGE/BERTScore ≠ clinical correctness; no clinician adjudication was performed.
- Challenge and safety sets are small; safety accuracy is optimistic.
- Local Rwandan brands absent from leaflets cannot be described beyond honest abstention.

- ZeroGPU free quota can interrupt anonymous demos; assessors should use a signed-in session.
- The assistant must not be used as medical advice; emergencies require SAMU 912 / a pharmacist.

AI assistance. AI coding tools helped scaffold code and draft challenge items; all training, evaluation, and live behaviour were run and verified by the author, who takes full responsibility for the submission. Commits are authored solely as Emmanuel-NS.

10. Conclusion

This project delivers a complete, reproducible domain-specific pharmacy QA system: trusted data, DistilBERT intent control, a fine-tuned MiniLM retriever, and a LoRA-adapted Qwen2.5 chat reader deployed on the web. Experiments show that **domain fine-tuning and consumer-style data** matter more than raw formal-test scores, that **LoRA roughly doubles** zero-shot Qwen answer quality while sharply improving groundedness and abstention, and that **live-user testing** surfaces failure modes (wrong labels, unknown brands, follow-ups) that offline benchmarks miss. Future work includes multilingual support, clinician evaluation, and expanding Rwanda leaflet coverage with OCR for scanned PDFs.

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Appendix A. Reproducibility checklist

- Clone https://github.com/Emmanuel-NS/FARUMASI-AI and install `requirements.txt`.
- Data: `python -m pharmqa.data.download && python -m pharmqa.data.prepare`.
- Full Colab pipeline: `notebooks/train_on_colab.ipynb`; chat reader: `notebooks/train_chat_reader_colab.ipynb`.
- Reported metrics are committed under `results/*.json`.
- Local UI: `python app.py`; Space deploy: `scripts/deploy_space.py`.

- Demo speaking outline for the required video: `report/DEMO_VIDEO_SCRIPT.md`.

Appendix B. How this report maps to the marking criteria

Criterion (pts)	Where addressed in this submission
Problem, language & dataset (8)	§1, §3; README data tables; dataset_stats.json
Model development (10)	§4; train_* scripts; explained again in the demo video
Baseline & experiments (8)	§5–§6; results/*.json; video must narrate why/what-learned
Evaluation & error analysis (7)	§6–§7; live success/fail cases; support & abstention
Research report (10)	This PDF: related work, method, results, discussion, refs
Web deployment (7)	§8; live Space URL; video workflow
Code & GitHub (5)	Repo layout, README reproduce, notebooks
Technical defense (5)	Demo video Q&A using DEMO_VIDEO_SCRIPT.md

Table 5. Submission map for assessors.

Supplementary materials: GitHub repository, Colab notebooks, results/.json, live Hugging Face Space, and this PDF under report/.*